SCIENCE 1.4 **Name:** ***ANSWERS***

WORKSHEET FIVE **REACTIONS OF ACIDS -Word equations**

1. **Key reactions to remember**. Complete the three general key reactions of acids.

ACID + BASE → ***SALT + WATER***

ACID + METAL → ***SALT + HYDROGEN***

ACID + CARBONATE → ***SALT + WATER + CARBON DIOXIDE***

2. **Acid + Metal**

(a) Sulfuric acid + magnesium → ***magnesium sulfate + hydrogen***

(b) Nitric acid + aluminium → ***aluminium nitrate + hydrogen***

(c) Hydrochloric acid + zinc → ***zinc chloride + hydrogen***

(d) Ethanoic acid + magnesium → ***magnesium ethanoate + hydrogen***

(e) sulfuric acid + zinc → ***zinc sulfate + hydrogen***

3. Acid + Carbonate

(a) Hydrochloric acid + sodium carbonate → ***sodium chloride + water + carbon dioxide***

(b) sulfuric acid + calcium carbonate → ***calcium sulfate + water + carbon dioxide***

(c) Nitric acid + potassium carbonate → ***potassium nitrate + water + carbon dioxide***

(d) Ethanoic acid + zinc carbonate → ***zinc ethanoate + water + carbon dioxide***

(e) Nitric acid + sodium bicarbonate → ***sodium nitrate + water + carbon dioxide***

4. **Acid + Base**

(a) Hydrochloric acid + sodium hydroxide → ***sodium chloride + water***

(b) Sulfuric acid + sodium hydroxide → ***sodium sulfate + water***

(c) Nitric acid + sodium hydroxide → ***sodium nitrate + water***

(d) Sulfuric acid + copper oxide → ***copper sulfate + water***

(e) Hydrochloric acid + copper oxide → ***copper chloride + water***

(f) Nitric acid + zinc oxide → ***zinc nitrate + water***

(g) Ethanoic acid + calcium oxide → ***calcium ethanoate + water***

(h) Hydrochloric acid + lead oxide → ***lead chloride + water***

(i) Sulfuric acid + copper hydroxide → ***copper sulfate + water***

(j) Ethanoic acid + iron(II) hydroxide → ***iron(II) ethanoate + water***

5. **Naming salts**

Name the salt formed when hydrochloric acid reacts with

(a) sodium hydroxide ***sodium chloride***

(b) copper oxide ***copper chloride***

(c) lithium carbonate ***lithium chloride***

(d) aluminium oxide ***aluminium chloride***

Name the salt formed when sulfuric acid reacts with

(a) calcium oxide ***calcium sulfate***

(b) lead carbonate ***lead sulfate***

(c) iron(III) hydroxide ***iron(III) sulfate***

(d) potassium hydroxide ***potassium sulfate***

6. **Identifying gases produced**

(a) Describe how you test for hydrogen gas and say what happens in a positive test.

Ignite the gas with a burning splint and it will burn with a "pop".

(b) Describe how you test for carbon dioxide gas and say what happens in a positive test.

Bubble the gas through limewater and the limewater will turn milky.